

UNIT I: GROWTH KINETICS & GROWTH PARAMETERS

Deep, exam-oriented notes for B.Sc. (Hons.) Biotechnology

Syllabus covered: microbial culture and growth kinetics; specific growth rate; doubling time; validity of exponential growth and exponential growth law; growth yield; metabolic quotient; mathematical model of simple batch culture; growth-associated kinetics; batch, fed-batch, cyclic fed-batch and continuous culture.

How to use these notes

This unit is easiest when learned as a chain:

Culture → **growth curve** → **mathematical model** → **growth parameters** → **reactor mode**.

Key idea

Microbial growth kinetics is the quantitative description of how cell mass, cell number, substrate and products change with time and with the culture environment. During balanced exponential growth, biomass behaves approximately as an autocatalytic process: $dX/dt = \mu X$.

Contents

1	Microbial Culture and Growth Kinetics	2
1.1	What is a microbial culture?	2
1.2	Why study growth kinetics?	2
1.3	Core variables	2
2	The Microbial Growth Curve	2
2.1	Lag phase	3
2.2	Exponential (log) phase	3
2.3	Stationary phase	3
2.4	Death or decline phase	3
3	Specific Growth Rate (μ)	3
3.1	Definition	3
3.2	Derivation for exponential growth	4
3.3	Connection with cell division	4
4	Doubling Time / Generation Time	4
4.1	Definition	4
4.2	Numerical example	4
5	Validity of Exponential Growth	5

5.1	Conditions supporting the model	5
5.2	How to test validity experimentally	5
6	Validity of the Exponential Growth Law	5
7	Growth Yield	6
7.1	Definition	6
7.2	Meaning	6
7.3	Relation between μ and substrate uptake	6
8	Metabolic Quotient	6
9	Mathematical Model of a Simple Batch Culture	7
9.1	Biomass balance	7
9.2	Substrate balance	7
9.3	A maintenance-inclusive model	7
10	Growth-Associated Kinetics	7
10.1	Growth-associated product formation	7
10.2	Luedeking-Piret model	8
11	Batch Culture	8
11.1	Definition	8
11.2	Typical profile	8
11.3	Advantages	8
11.4	Limitations	8
12	Fed-Batch Culture	8
12.1	Definition	9
12.2	Why use fed-batch?	9
12.3	Common feeding strategies	9
13	Cyclic Fed-Batch Culture	9
13.1	Concept	9
13.2	Potential advantages	10
14	Continuous Culture	10
14.1	Definition	10
14.2	Dilution rate	10
14.3	Steady-state substrate relation	11
14.4	Washout	11
15	Batch vs Fed-Batch vs Cyclic Fed-Batch vs Continuous	11

16 Monod Kinetics: A High-Yield Extension	11
17 Worked Numerical Problems	12
18 Practical Data Analysis Workflow	12
19 Rapid Revision Sheet	13
20 Exam Questions You Should Be Able to Answer	13
21 Reference Notes	14

1 Microbial Culture and Growth Kinetics

1.1 What is a microbial culture?

A microbial culture is a population of microorganisms grown under controlled nutritional and physicochemical conditions. A useful culture system contains an inoculum, a growth medium, a vessel or bioreactor, and defined operating conditions such as temperature, pH, oxygen transfer, mixing and nutrient supply.

1.2 Why study growth kinetics?

Growth kinetics connects laboratory measurements to biological interpretation and process design. It helps answer questions such as: How quickly is biomass increasing? Which nutrient limits growth? How much biomass can be produced per gram of substrate? When should harvesting occur? What happens when fresh medium is continuously supplied?

Biological meaning

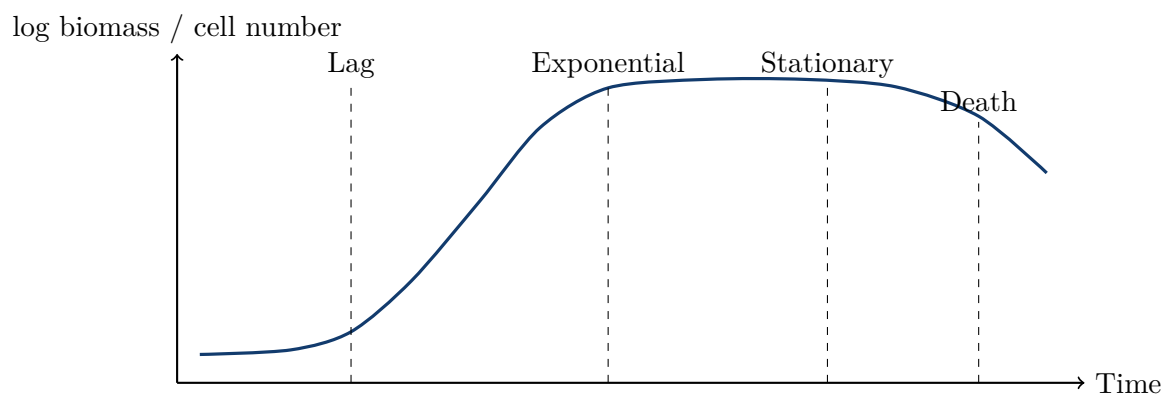
The same culture may behave very differently when oxygen, carbon source, temperature, pH, osmotic pressure, or nutrient balance changes. Therefore, growth rate is not a fixed species number; it is a property of the organism under a defined set of conditions.

1.3 Core variables

Symbol	Meaning
X	Biomass concentration, e.g. g L ⁻¹ dry weight
N	Cell number or population size
S	Concentration of growth-limiting substrate
P	Product concentration
μ	Specific growth rate (time ⁻¹)
$\mu_{\max}$	Maximum specific growth rate under the stated conditions
$Y_{X/S}$	Biomass yield on substrate, biomass formed per substrate consumed
q_S	Specific substrate uptake rate
D	Dilution rate in continuous culture
F	Volumetric feed rate
V	Culture volume

2 The Microbial Growth Curve

A closed batch culture usually passes through lag, exponential, stationary and death/decline phases. The exact shape depends on organism and conditions.



2.1 Lag phase

Cells adapt to the physical and nutritional environment. Metabolic reprogramming, repair, synthesis of enzymes and macromolecules, and adaptation to stress may occur. Cell division is usually limited compared with exponential phase.

2.2 Exponential (log) phase

Cells divide at approximately constant specific growth rate. If biomass is plotted on a logarithmic scale, the exponential phase becomes a straight line. The slope of the natural-log biomass plot is μ .

2.3 Stationary phase

Net growth approaches zero because nutrient limitation, oxygen limitation, toxic product accumulation, pH changes, or other stresses balance cell formation. Cells can remain metabolically active even when net biomass no longer rises.

2.4 Death or decline phase

Viable-cell number can decrease. Death may result from starvation, toxin accumulation, lysis, or severe environmental stress.

Exam focus

In most calculations of μ and doubling time, choose data points from the exponential phase only. Mixing lag or stationary-phase points into a log-linear regression gives a misleading growth rate.

3 Specific Growth Rate (μ)

3.1 Definition

Specific growth rate is the increase in biomass per unit biomass per unit time:

$$\mu = \frac{1}{X} \frac{dX}{dt}.$$

Therefore,

$$\frac{dX}{dt} = \mu X.$$

Units are usually h^{-1} or min^{-1} .

3.2 Derivation for exponential growth

For constant μ :

$$\frac{dX}{X} = \mu dt$$

Integrating from X_0 at $t = 0$ to X_t at time t :

$$\ln\left(\frac{X_t}{X_0}\right) = \mu t$$

so

$$X_t = X_0 e^{\mu t}.$$

This is the exponential growth law.

3.3 Connection with cell division

For ideal binary fission, if one generation takes time g , then after n generations:

$$X = X_0 2^n.$$

Since $n = t/g$,

$$X = X_0 2^{t/g} = X_0 e^{(\ln 2)t/g}.$$

Comparing with $X = X_0 e^{\mu t}$ gives

$$\mu = \frac{\ln 2}{g}.$$

Hence $g = \ln 2 / \mu$.

Key idea

Interpretation: μ answers “how fast is the biomass increasing relative to how much biomass is already present?” A larger μ means faster proportional growth.

4 Doubling Time / Generation Time

4.1 Definition

Doubling time t_d is the time required for biomass or cell number to increase twofold during exponential growth.

$$X(t + t_d) = 2X(t).$$

Using $X = X_0 e^{\mu t}$:

$$2 = e^{\mu t_d} \Rightarrow t_d = \frac{\ln 2}{\mu}.$$

4.2 Numerical example

If $\mu = 0.35 \text{ h}^{-1}$:

$$t_d = \frac{0.693}{0.35} = 1.98 \text{ h}.$$

Thus the biomass doubles approximately every 2 h during the exponential period.

Exam focus

Fastest route in numericals: first calculate $\ln(X_2/X_1)/\Delta t = \mu$, then use $t_d = 0.693/\mu$.

5 Validity of Exponential Growth

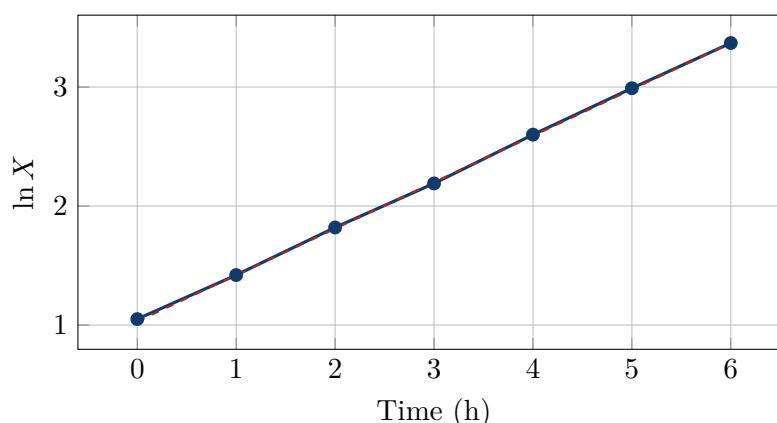
The exponential model is useful but conditional. It assumes that the specific growth rate is approximately constant over the selected time interval.

5.1 Conditions supporting the model

1. Cells are in the exponential phase.
2. The growth environment is sufficiently stable.
3. The relevant nutrient(s) are available at concentrations that permit approximately constant μ .
4. Temperature and pH remain near controlled values.
5. Oxygen transfer is not changing enough to impose a new limitation.
6. Toxic metabolites do not accumulate to a level that changes growth kinetics.
7. The measurement of biomass or optical density is reliable and comparable across time.

5.2 How to test validity experimentally

Plot $\ln X$ versus time. If the points form an approximately straight line over a defined interval, exponential growth is a reasonable approximation. The slope gives μ .



Common mistake

A straight line on a regular (linear) biomass-vs-time plot is NOT the signature of exponential growth. Exponential growth appears curved on ordinary axes but becomes linear when the biomass axis is logarithmic (or when $\ln X$ is plotted directly).

6 Validity of the Exponential Growth Law

The equation $X_t = X_0 e^{\mu t}$ is an integrated form of $dX/dt = \mu X$ under constant μ . Therefore it fails when μ changes substantially with time.

A practical workflow is:

1. identify the likely exponential region;
2. calculate $\ln X$ for the biomass measurements;

3. plot $\ln X$ against time;
4. estimate slope and goodness of fit;
5. check that residuals do not show systematic curvature;
6. report the time interval used for the calculation.

7 Growth Yield

7.1 Definition

Growth yield quantifies how efficiently substrate is converted into biomass. The biomass yield coefficient is:

$$Y_{X/S} = \frac{\Delta X}{-\Delta S}$$

when substrate concentration decreases as biomass increases. Typical units are g biomass per g substrate.

7.2 Meaning

A high $Y_{X/S}$ means more biomass is formed per unit substrate consumed. Yield depends on the organism, substrate, oxygen availability, maintenance requirements, by-product formation, and measurement definitions.

7.3 Relation between μ and substrate uptake

Define the specific substrate uptake rate q_S as substrate consumed per unit biomass per unit time. For a simple growth-associated conversion:

$$\mu = Y_{X/S} q_S.$$

Thus,

$$q_S = \frac{\mu}{Y_{X/S}}.$$

This relationship is most directly useful when substrate consumption is predominantly associated with biomass synthesis.

8 Metabolic Quotient

The term metabolic quotient is used for a rate normalized to biomass, but its exact definition depends on the physiological context. In microbial kinetics, a common and useful form is the specific substrate consumption/uptake rate:

$$q_S = \frac{1}{X} \left(-\frac{dS}{dt} \right).$$

For a respiratory substrate such as oxygen, a corresponding specific oxygen uptake rate can be defined:

$$q_{O_2} = \frac{1}{X} \left(-\frac{dC_{O_2}}{dt} \right).$$

Biological meaning

The key idea is normalization by biomass. If 10 mg substrate is consumed per hour, that number means something very different in a 0.1 g L⁻¹ culture than in a 10 g L⁻¹ culture. Specific rates remove much of this scale dependence.

9 Mathematical Model of a Simple Batch Culture

A simple batch system is closed with respect to material addition/removal during the growth experiment (apart from gas exchange and sampling, depending on the setup).

9.1 Biomass balance

For growth-associated biomass formation:

$$\frac{dX}{dt} = \mu X.$$

Under constant μ :

$$X = X_0 e^{\mu t}.$$

9.2 Substrate balance

Let substrate uptake be described by q_S :

$$\frac{dS}{dt} = -q_S X.$$

For purely growth-associated uptake where $\mu = Y_{X/S} q_S$:

$$\frac{dS}{dt} = -\frac{\mu X}{Y_{X/S}}.$$

Then:

$$\frac{dX}{dS} = -Y_{X/S},$$

and integration gives the simple linear yield relation:

$$\boxed{X - X_0 = Y_{X/S}(S_0 - S)}.$$

9.3 A maintenance-inclusive model

A more realistic model may include substrate consumed for non-growth maintenance:

$$q_S = \frac{\mu}{Y_G} + m_S,$$

where Y_G is a growth yield coefficient and m_S is a maintenance coefficient. Then substrate consumption is not perfectly proportional to biomass formation.

10 Growth-Associated Kinetics

10.1 Growth-associated product formation

In growth-associated product formation, the product formation rate is proportional to biomass growth:

$$\frac{dP}{dt} = \alpha \mu X.$$

If biomass concentration is increasing by $dX/dt = \mu X$, then:

$$\frac{dP}{dX} = \alpha.$$

So the idealized integrated relation is:

$$P - P_0 = \alpha(X - X_0).$$

10.2 Luedeking-Piret model

A general model for product formation is:

$$\frac{dP}{dt} = \alpha\mu X + \beta X$$

where α is the growth-associated coefficient and β is the non-growth-associated coefficient.

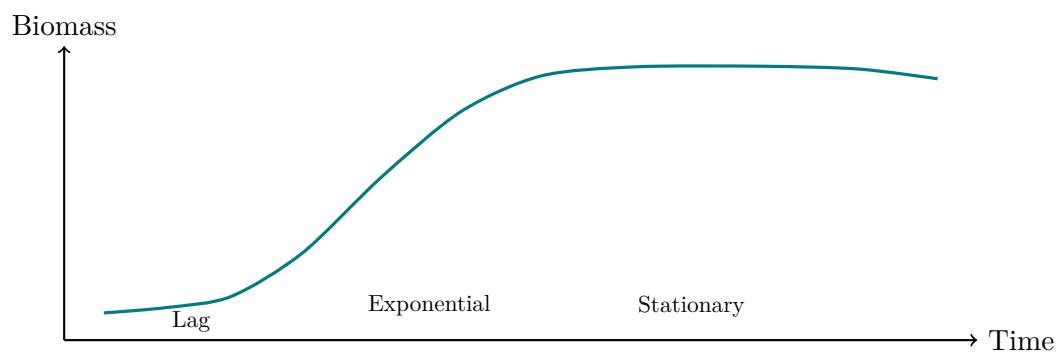
Kinetic type	Mathematical signature	Typical interpretation
Growth-associated	$\alpha > 0, \beta \approx 0$	Product rises mainly while biomass grows
Non-growth-associated	$\alpha \approx 0, \beta > 0$	Product can form during stationary phase
Mixed	$\alpha > 0, \beta > 0$	Product forms during growth and persists/continues later

11 Batch Culture

11.1 Definition

Batch culture is a closed cultivation mode in which inoculum and medium are introduced at the start and the culture is allowed to proceed without continuous nutrient feed or continuous culture withdrawal.

11.2 Typical profile



11.3 Advantages

Simple operation; no feed pump is required; suitable for inoculum preparation and basic kinetics; easy to collect a complete time course.

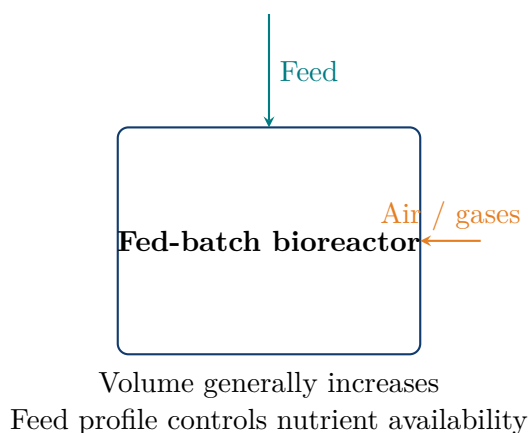
11.4 Limitations

Nutrients change continuously; inhibitory products may accumulate; physiological state changes over time; maximum exponential growth cannot be maintained indefinitely.

12 Fed-Batch Culture

12.1 Definition

Fed-batch is a semi-closed cultivation mode in which one or more nutrients are supplied during cultivation while culture volume usually increases. There is typically no continuous culture effluent during the main feeding phase.



12.2 Why use fed-batch?

Fed-batch operation can extend the period of productive growth, prevent rapid exhaustion of a key substrate, reduce overflow metabolism in some systems, and enable high cell density by controlling feed rate.

12.3 Common feeding strategies

Constant feed, step feed, exponential feed and feedback-controlled feed. An exponential feed can be designed to approximate a target specific growth rate during a period of biomass accumulation.

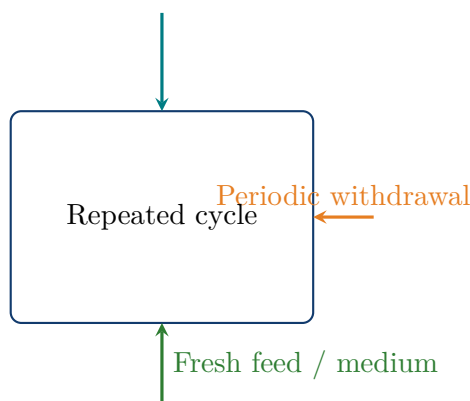
Common mistake

Fed-batch is not the same as continuous culture. In fed-batch, feed enters but routine culture effluent is absent or limited, so volume changes and cells are not continuously replaced by an equal outflow.

13 Cyclic Fed-Batch Culture

13.1 Concept

Cyclic fed-batch combines repeated feeding with periodic withdrawal/harvesting of part of the culture. Fresh medium or concentrated feed is then supplied to restore operating conditions. Depending on the design, this can combine aspects of batch and repeated fill-and-draw operation.



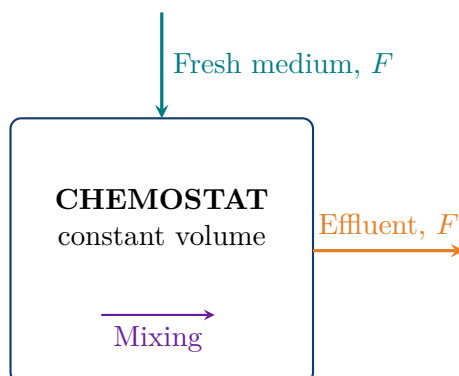
13.2 Potential advantages

It can prolong cultivation, partially reset substrate and product conditions, and support repeated biomass or product harvests without fully restarting from a small inoculum each time.

14 Continuous Culture

14.1 Definition

Continuous culture maintains a culture at approximately constant volume by continuously adding fresh medium and removing culture at the same rate. A common implementation is the chemostat.



14.2 Dilution rate

The dilution rate is

$$D = \frac{F}{V} \quad (\text{time}^{-1}).$$

At ideal steady state, biomass balance gives:

$$0 = \mu X - DX,$$

so for non-zero biomass:

$$\mu = D.$$

This is one of the most important equations in continuous culture.

14.3 Steady-state substrate relation

For a chemostat controlled by one limiting substrate, the steady-state condition connects $\mu = D$ to the kinetic model for $\mu(S)$. With the Monod equation:

$$\mu = \mu_{\max} \frac{S}{K_S + S}.$$

At steady state,

$$D = \mu_{\max} \frac{S}{K_S + S}.$$

Solving:

$$S = \frac{DK_S}{\mu_{\max} - D}.$$

The equation shows why D approaching $\mu_{\max}$ makes the predicted residual substrate rise sharply.

14.4 Washout

When the dilution rate is increased beyond the culture's achievable growth rate, cells cannot replace those leaving the vessel. Biomass concentration falls and the culture can wash out. In idealized theory, washout occurs near $D \geq \mu_{\max}$, but real systems may depart from the ideal assumption because of adaptation, multiple limitations, biomass retention, and other effects.

Exam focus

At steady state in an ideal chemostat: $\mu = D$. Doubling time can therefore be written as $t_d = \ln 2/D$ when the culture is truly at steady state.

15 Batch vs Fed-Batch vs Cyclic Fed-Batch vs Continuous

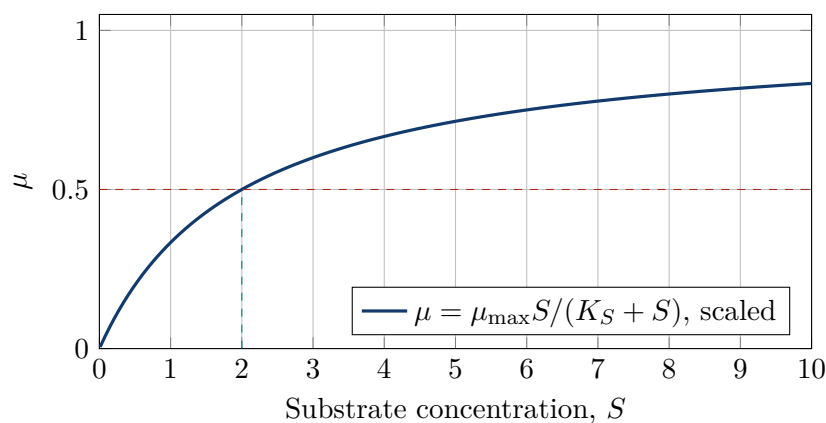
Mode	Feed?	Effluent?	Volume	Main kinetic character
Batch	No routine feed	No routine outflow	Nearly constant	Environment changes with time
Fed-batch	Yes	Usually no continuous outflow	Increases	Nutrient supply is controlled over time
Cyclic fed-batch	Repeated feed	Periodic withdrawal	Cycles	Repeated fill/withdraw operation
Continuous	Yes, continuously	Yes, continuously	Constant (ideal)	Steady state possible; $\mu = D$

16 Monod Kinetics: A High-Yield Extension

Although not explicitly listed as a separate line in the supplied syllabus image, Monod kinetics provides the conceptual bridge between substrate concentration and specific growth rate and is widely used when interpreting microbial growth kinetics.

$$\mu = \mu_{\max} \frac{S}{K_S + S}$$

Here K_S is the substrate concentration at which $\mu = \mu_{\max}/2$ in the Monod model.



17 Worked Numerical Problems

Problem 1: Specific growth rate

A culture increases from $X_1 = 0.20 \text{ g L}^{-1}$ to $X_2 = 1.00 \text{ g L}^{-1}$ in 4 h during exponential growth.

$$\mu = \frac{\ln(1.00/0.20)}{4} = \frac{\ln 5}{4} = 0.402 \text{ h}^{-1}.$$

Answer: $\mu \approx 0.402 \text{ h}^{-1}$.

Problem 2: Doubling time

For $\mu = 0.402 \text{ h}^{-1}$,

$$t_d = \frac{0.693}{0.402} = 1.72 \text{ h}.$$

Problem 3: Yield

Suppose biomass rises from 0.5 to 2.5 g L^{-1} while substrate falls from 20 to 12 g L^{-1} .

$$Y_{X/S} = \frac{2.5 - 0.5}{20 - 12} = \frac{2}{8} = 0.25 \text{ g/g}.$$

Problem 4: Continuous culture

A chemostat has $F = 0.6 \text{ L h}^{-1}$ and $V = 3 \text{ L}$.

$$D = F/V = 0.6/3 = 0.20 \text{ h}^{-1}.$$

At ideal steady state, $\mu = 0.20 \text{ h}^{-1}$ and

$$t_d = 0.693/0.20 = 3.47 \text{ h}.$$

18 Practical Data Analysis Workflow

For your own OD-time experiment, use the following sequence:

1. Record time and OD consistently.
2. Blank-correct OD if appropriate.

3. Keep the instrument within its linear measurement range; dilute dense samples when needed.
4. Identify the exponential window from the growth curve.
5. Plot $\ln(\text{OD})$ versus time for the chosen window.
6. Fit a straight line; the slope is an estimate of μ if OD is proportional to biomass in that range.
7. Compute $t_d = 0.693/\mu$.
8. Do not calculate μ from lag or stationary phase points.
9. Report temperature, medium, aeration, shaking/mixing and inoculum conditions with the kinetic value.

Common mistake

OD is an indirect proxy for biomass. At high cell density, light scattering becomes non-linear; pigments, cell size, aggregation and instrument configuration can also affect the OD-biomass relationship. Therefore, do not automatically interpret raw OD as grams of biomass without calibration.

19 Rapid Revision Sheet

Concept	Formula / key statement
Specific growth rate	$\mu = (1/X)(dX/dt)$
Exponential growth	$X = X_0 e^{\mu t}$
Doubling time	$t_d = \ln 2/\mu$
Generation number	$n = t/t_d$
Yield coefficient	$Y_{X/S} = \Delta X/(-\Delta S)$
Specific substrate uptake	$q_S = (1/X)(-dS/dt)$
Growth-associated product	$dP/dt = \alpha\mu X$
Luedeking-Piret	$dP/dt = \alpha\mu X + \beta X$
Dilution rate	$D = F/V$
Chemostat steady state	$\mu = D$
Monod model	$\mu = \mu_{\max} S/(K_S + S)$
Ideal washout condition	D approaching/exceeding the sustainable growth rate

20 Exam Questions You Should Be Able to Answer

1. Define microbial growth kinetics and specific growth rate.
2. Derive the exponential growth equation from $dX/dt = \mu X$.
3. Derive the relationship between specific growth rate and doubling time.
4. Explain the four phases of a microbial growth curve with a labelled diagram.
5. State the assumptions and limitations of exponential growth.
6. Define growth yield and derive its relationship with substrate uptake.
7. Explain metabolic quotient/specific uptake rate and its biological importance.
8. Develop a simple batch-culture mass balance for biomass and substrate.
9. Explain growth-associated kinetics and the Luedeking-Piret model.
10. Differentiate batch, fed-batch, cyclic fed-batch and continuous culture.

11. Derive $\mu = D$ at chemostat steady state.
12. Explain dilution rate and washout.
13. Explain why a chemostat can maintain cells at a reproducible physiological state.

21 Reference Notes

These notes are synthesized from the supplied syllabus and standard microbial-growth kinetics concepts. Contemporary open-access reviews emphasize the central role of μ , exponential growth analysis, batch versus continuous cultivation, and chemostat steady-state relationships. See, for example, Kovarova-Kovar and Egli on growth kinetics, the Microbiology Society primer on bacterial growth kinetics, and NCBI's overview of continuous-culture chemostat systems.

Web references consulted: Kovarova-Kovar & Egli, *Growth Kinetics of Suspended Microbial Cells*; Microbial Primer: Bacterial Growth Kinetics (2024); NCBI Bookshelf, Continuous-Culture Chemostat Systems.